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Unit 3 - Programming the Butterfly Display

PDF Documents

PDF Document 1: 1738216888367-Chapter 1 - Unit 3 - Programming the Butterfly Display.pdf

This PDF document is included in the unit materials.

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UNIT 3 PROGRAMMING THE BUTTERFLY DISPLAY

Let's create a butterfly flight simulator!

Get ready to become a butterfly pilot!

In this exciting activity, we'll learn how to program a computer to make a butterfly fly around on the screen. We'll use a joystick to control the butterfly's movements, just like a real pilot!



WHAT WILL WE DO?

1 BECOME A CODING WIZARD

We'll learn special commands that tell the computer what to do. These commands will help us make the butterfly fly.

2 CREATE OUR OWN BUTTERFLY

We'll put our own beautiful butterfly on the Display. We can choose the color, pattern, and even add sparkles!

3 CONTROL THE BUTTERFLY WITH A JOYSTICK

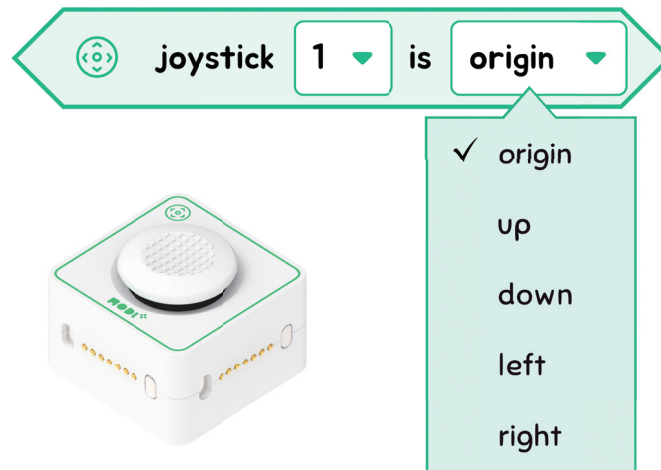
Using a joystick, we'll be able to move the butterfly up, down, left, and right, just like a real pilot flying a plane.

It's easy! With a little help, everyone can create their own flying butterfly. Are you ready to start this fun coding adventure?

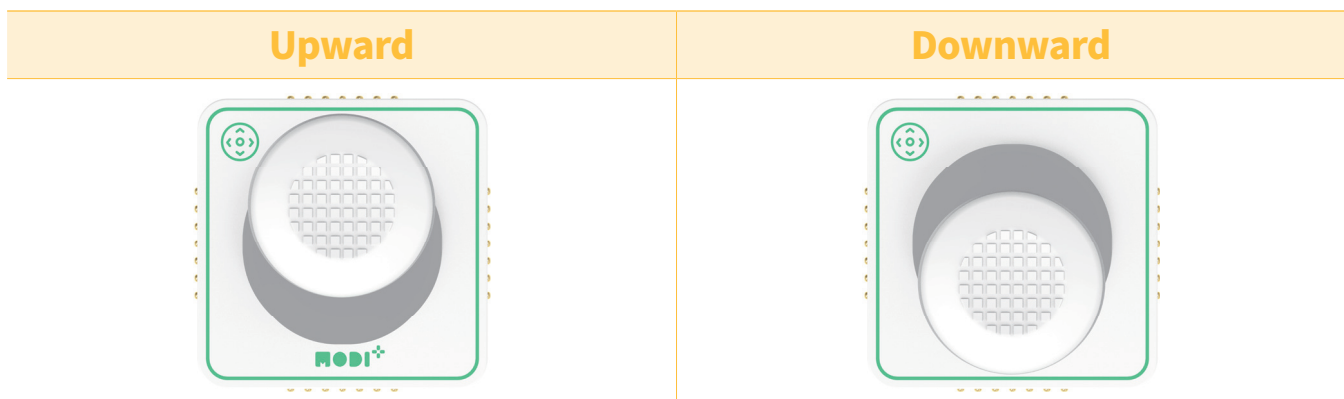


Module Functionality

Let's look into the details of the Joystick module.

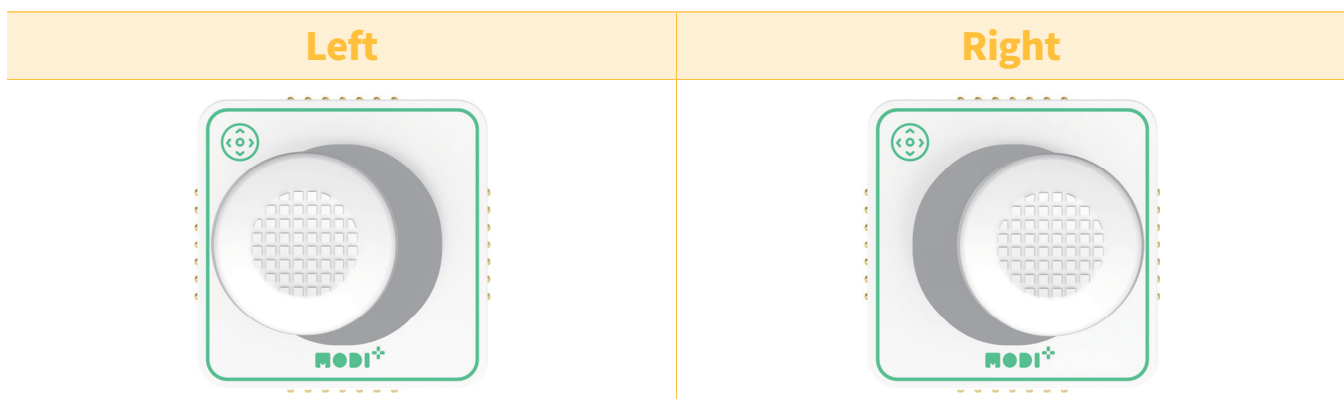


Including the neutral position 'origin,' the joystick has 5 different positions.



Pushing it upward and downward affects movement on the **y-axis**

- Range : -100 % (fully downward) ~ +100 % (fully upward)

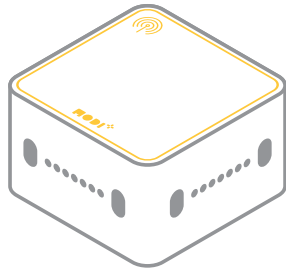


Moving the joystick left and right controls movement along the **x-axis**

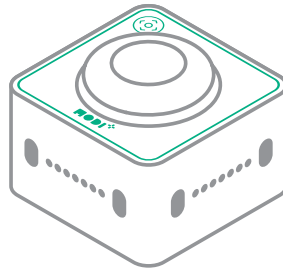
- Range : -100 % (fully left) ~ +100 % (fully right)



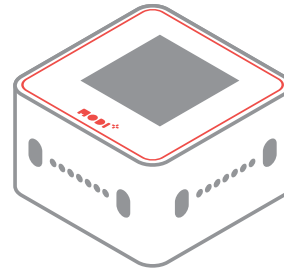
Material Needed



Newtork



Joystick



Display

When we combine all the modules, a creation has to look like this.



Code Story

Show butterfly on Display



When Joystick is up



Move Display up by 3px



When Joystick is right



Move Display right by 3px

When Joystick is down



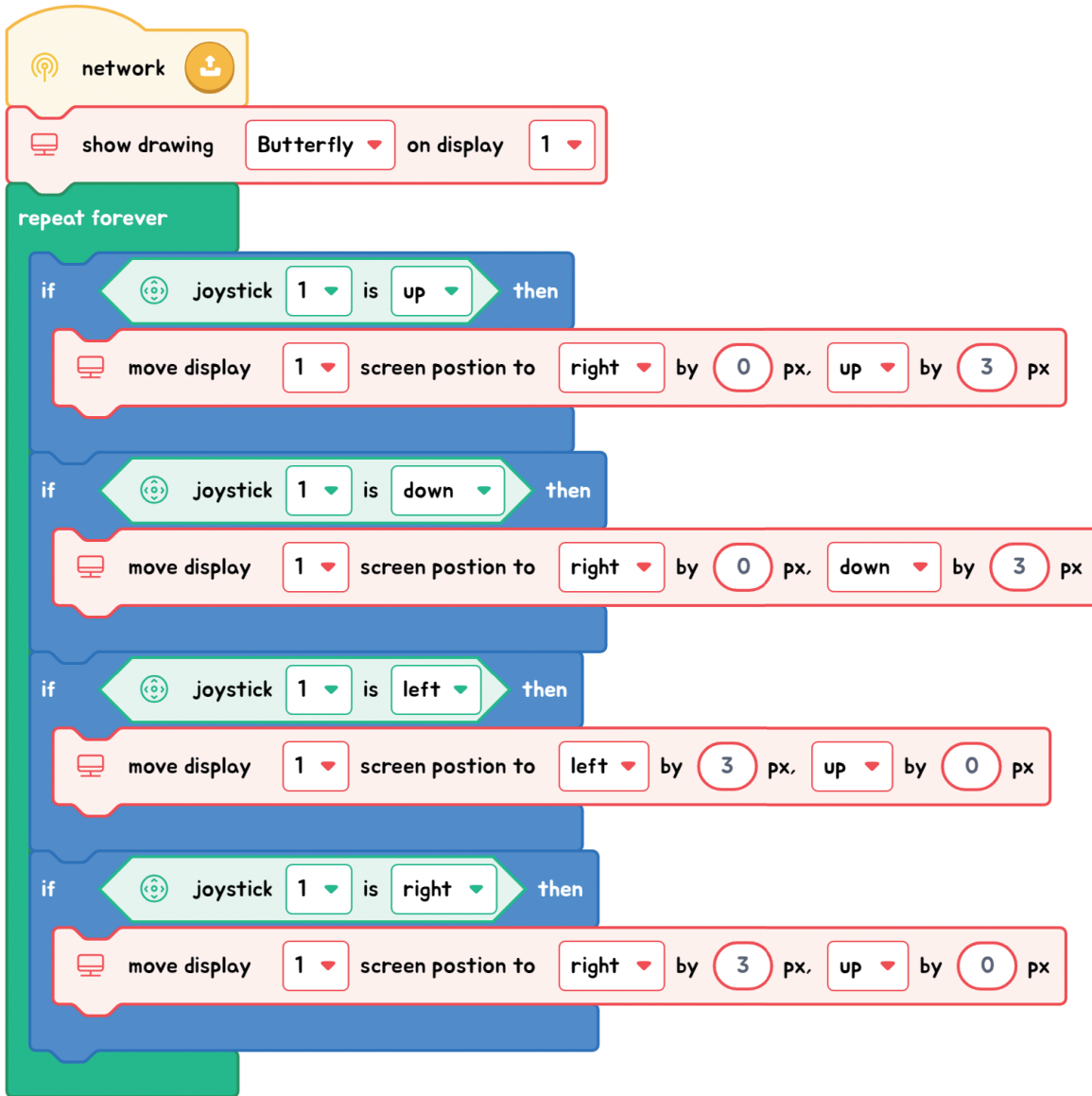
Move Display down by 3px

When Joystick is left



Move Display left by 3px

Code Quest



Checkpoint

Let's add code that play a sound while it moves forward and backward.

